$= 8.23 = 8.2$ m

7 **C** By principle of conservation of momentum, take direction to the right as positive for the block of mass m

$$mu - 2m(0.5u) = 3mv$$
$$v = 0$$

By principle of conservation of energy,

$\frac{1}{2}mu^2 + \frac{1}{2}(2m)(0.5u)^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$, where $v = 0$

$$x = \sqrt{\frac{3mu^2}{2k}}$$

8 **A** θ is the □ the string makes with the vertical